Wastewater Microbial Indicators and Pathogens: Prevalence, Removal, and Surrogate Assessment

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1. Introduction

This report summarizes indicator microorganisms and pathogenic targets measured across three wastewater treatment plants between March 2015 and June 2016. The dataset includes bacterial indicators (total coliform, fecal coliform, E. coli), bacteriophages (somatic and male-specific coliphages), aerobic endospores, protozoan parasites (Giardia and Cryptosporidium), and adenovirus. Results are interpreted in terms of detection prevalence and log-removal values (LRVs) to support treatment performance understanding and surrogate selection for pathogen control.

2. Objectives

The assessment focuses on three questions that directly support process performance evaluation and monitoring strategy development:

1. Quantify prevalence of indicator and pathogen detections in influent and effluent across plants.
2. Evaluate microbial removal through treatment (LRVs), including plant-to-plant comparisons.
3. Assess the extent to which indicator LRVs track pathogen LRVs (surrogate effectiveness) when plants are pooled.

3. Description of Dataset

The source workbook (WWTP_microbial_loads_and_removal.xlsx) contains event-based measurements by plant, sampling month, sample type (influent grab, effluent grab, UF effluent), and microorganism target. Non-detects are represented as BDL or numeric zero; these were treated as method detection limit (MDL) values in the underlying analysis. For protozoa and coliphages, the UF effluent sample was used as the effluent representation; for all other targets, the effluent grab sample was used.

The three plants and their major treatment and disinfection configurations are summarized below to support interpretation of plant-to-plant differences:

Table 1 – Wastewater treatment plants included in the dataset and their disinfection configurations.

Plant (short name)	Plant (full name)	Primary/secondary treatment train	Disinfection
NoWe	Northwest El Paso	Primary clarification, activated sludge, secondary clarification, disc filter microscreens	UV
BiSp	Big Spring	Primary clarification, activated sludge/trickling filter, secondary clarification	Chlorine
BrWo	Brownwood	Primary clarification, activated sludge, secondary clarification	Chlorine

Plant short names used throughout are: NoWe (Northwest El Paso), BiSp (Big Spring), and BrWo (Brownwood).

4. Analysis

4.1 Detection prevalence (influent vs effluent)

Detection prevalence provides a first-pass view of how consistently each target occurs and whether effluent results are frequently censored by the MDL. Tables 2 and 3 summarize pooled detection prevalence across all plants. Full by-plant prevalence tables are provided in the Appendix (Tables 8 and 9).

Table 2 – Influent detection prevalence (all plants pooled).

Microorganism	n	Detections	Prevalence (%)
Adenovirus	48	46	95.8
Aerobic endospores	45	45	100.0
Cryptosporidium	47	31	66.0
E. coli	45	45	100.0
Fecal coliform	45	45	100.0
Giardia	47	46	97.9
Male-specific coliphage	45	45	100.0
Somatic coliphage	44	44	100.0
Total coliform	42	42	100.0

Table 3 – Effluent detection prevalence (all plants pooled).

Microorganism	n	Detections	Prevalence (%)
Adenovirus	48	28	58.3
Aerobic endospores	45	45	100.0
Cryptosporidium	47	39	83.0
E. coli	44	31	70.5
Fecal coliform	44	33	75.0
Giardia	47	45	95.7
Male-specific coliphage	46	5	10.9
Somatic coliphage	46	9	19.6
Total coliform	41	39	95.1

Across all plants, influent detections are near-universal for bacterial indicators and both coliphage groups, whereas influent Cryptosporidium prevalence is materially lower than other targets. In effluent, coliphages show frequent non-detects while aerobic endospores remain consistently detected, indicating persistence through the treatment train and limited sensitivity to disinfection.

4.2 Influent and effluent concentration statistics

Concentration summaries in original units provide context for interpreting LRVs, particularly where effluent values are censored at the MDL. Tables 4 and 5 present pooled influent and effluent statistics. Full by-plant concentration statistics are provided in the Appendix (Tables 10 and 11).

Table 4 – Influent concentrations summary (all plants pooled; original units).

Microorganism	n	Median	Mean	P25	P75

Adenovirus	48	107	4,541	17	5,819
Aerobic endospores	45	210	568	110	445
Cryptosporidium	47	10	269	0	50
E. coli	45	7,120,000	82,054,222	4,000,000	53,400,000
Fecal coliform	45	17,900,000	188,166,444	6,200,000	137,000,000
Giardia	47	410	2,078	120	2,510
Male-specific coliphage	45	2,310	3,003	1,060	5,000
Somatic coliphage	44	1,430	2,174	906	2,784
Total coliform	42	83,900,000	426,861,190	25,732,500	327,650,000

Table 5 – Effluent concentrations summary (all plants pooled; original units).

Microorganism	n	Median	Mean	P25	P75
Adenovirus	48	0	15	0	0
Aerobic endospores	45	8	19	4	18
Cryptosporidium	47	0	2	0	1
E. coli	44	2	16	1	9
Fecal coliform	44	7	52	1	46
Giardia	47	4	15	3	20
Male-specific coliphage	46	1	6	1	1
Somatic coliphage	46	1	7	1	1
Total coliform	41	345	673	68	1,120

Pooled results indicate that bacterial indicators occur at very high influent levels (orders of magnitude higher than protozoa and adenovirus). Effluent distributions for coliphages are strongly influenced by MDL censoring (median effluent values at or near the MDL), whereas adenovirus shows evidence of episodic breakthrough in some events (reflected by a high effluent mean relative to the median). These patterns are consistent with strong baseline treatment performance punctuated by event-driven variability that is more apparent for pathogens than for bacterial indicators.

4.3 Log-removal values by plant

Figures 1 to 3 compare LRV distributions by plant, separated into indicator organisms, protozoan parasites, and adenovirus. Markers differentiate events where the effluent was non-detect (MDL-censored) versus detected, which is important for interpreting upper tails that may be driven by substitution at low effluent values.

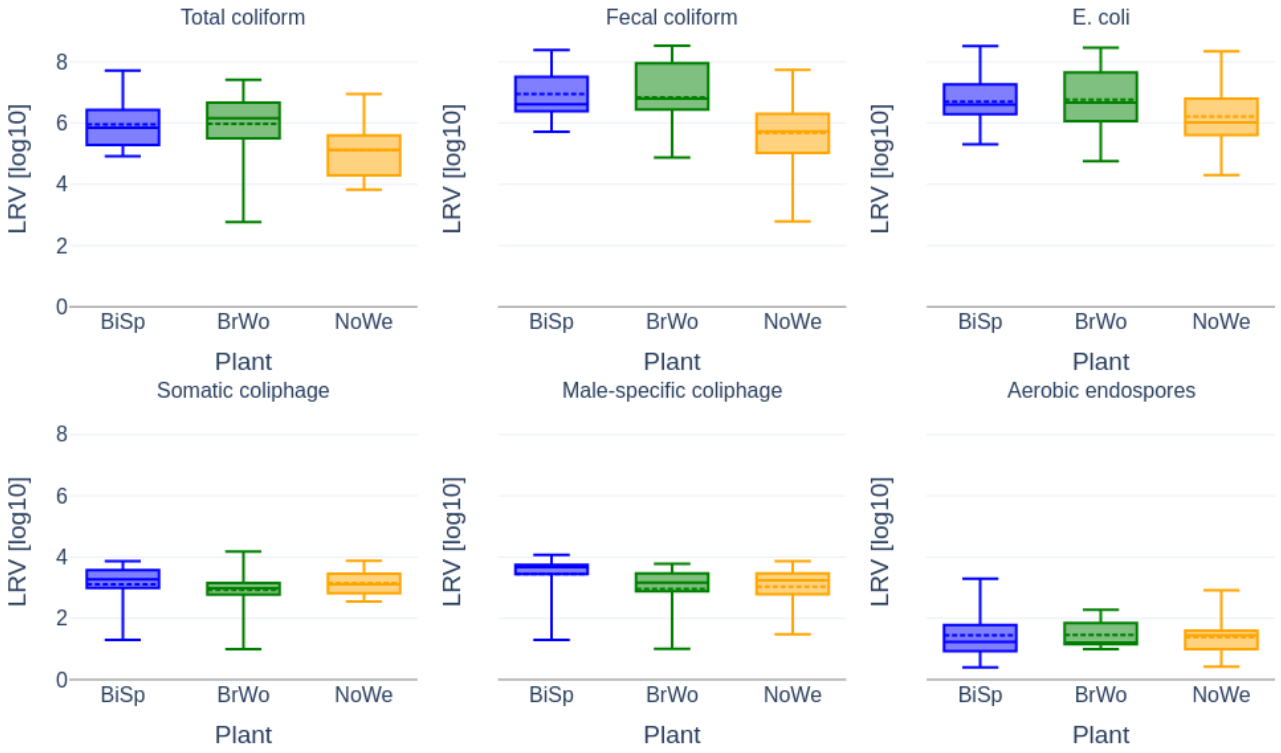

Figure 1 – Indicator microorganism LRVs by plant (non-detect and detect effluent markers).

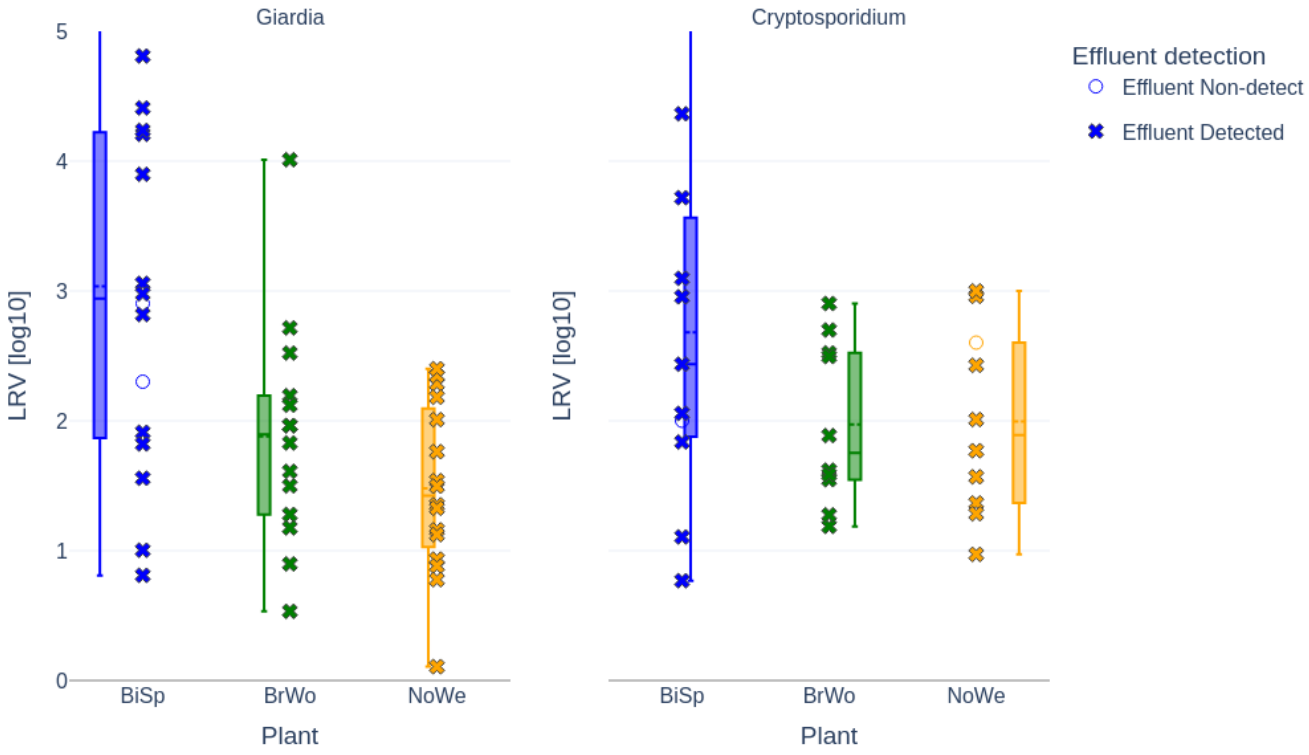

Figure 2 – Protozoan parasite LRVs by plant (non-detect and detect effluent markers).

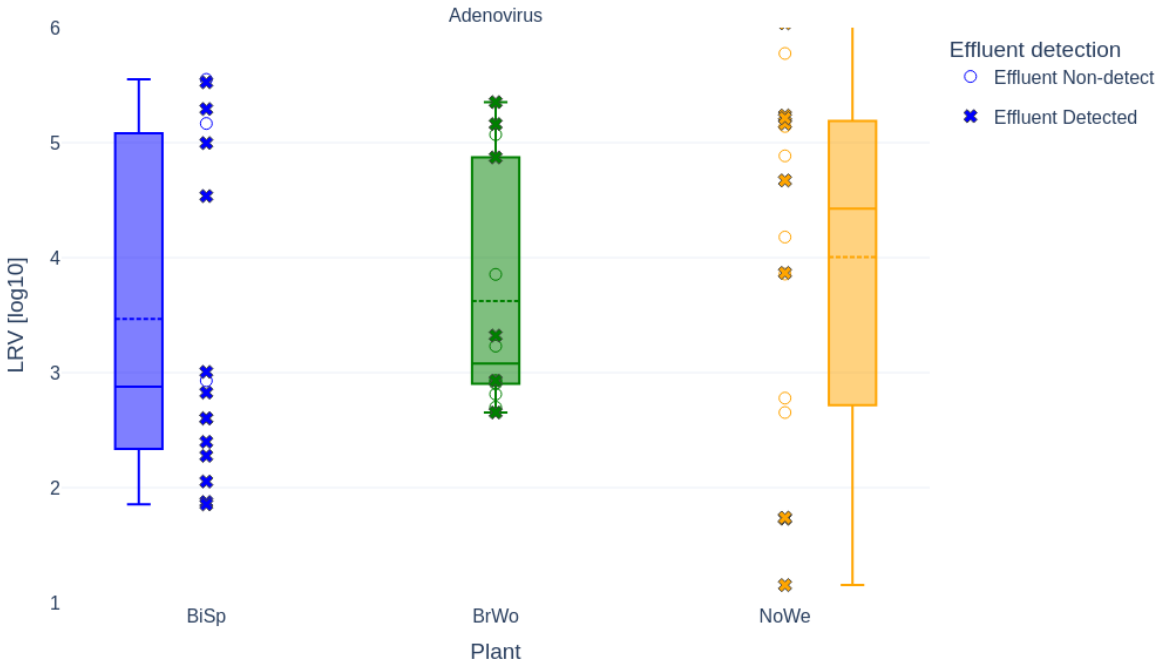

Figure 3 – Adenovirus LRVs by plant (non-detect and detect effluent markers).

Bacterial indicators exhibit consistently high LRVs (typically around 5 to 7 log10), indicating strong reduction through the combined primary and secondary trains and disinfection. In contrast, protozoa show substantially lower and more variable removal, consistent with greater reliance on physical separation and solids/particle capture rather than disinfection. Adenovirus exhibits wide within-plant spread, consistent with disinfection sensitivity and episodic performance excursions.

4.4 Pooled LRV patterns and summary statistics

Figure 4 shows LRVs pooled across plants, highlighting distinct LRV bands by microorganism group. Table 6 summarizes pooled LRV statistics (median, mean, and interquartile range). Full by-plant LRV statistics are provided in the Appendix (Table 12).

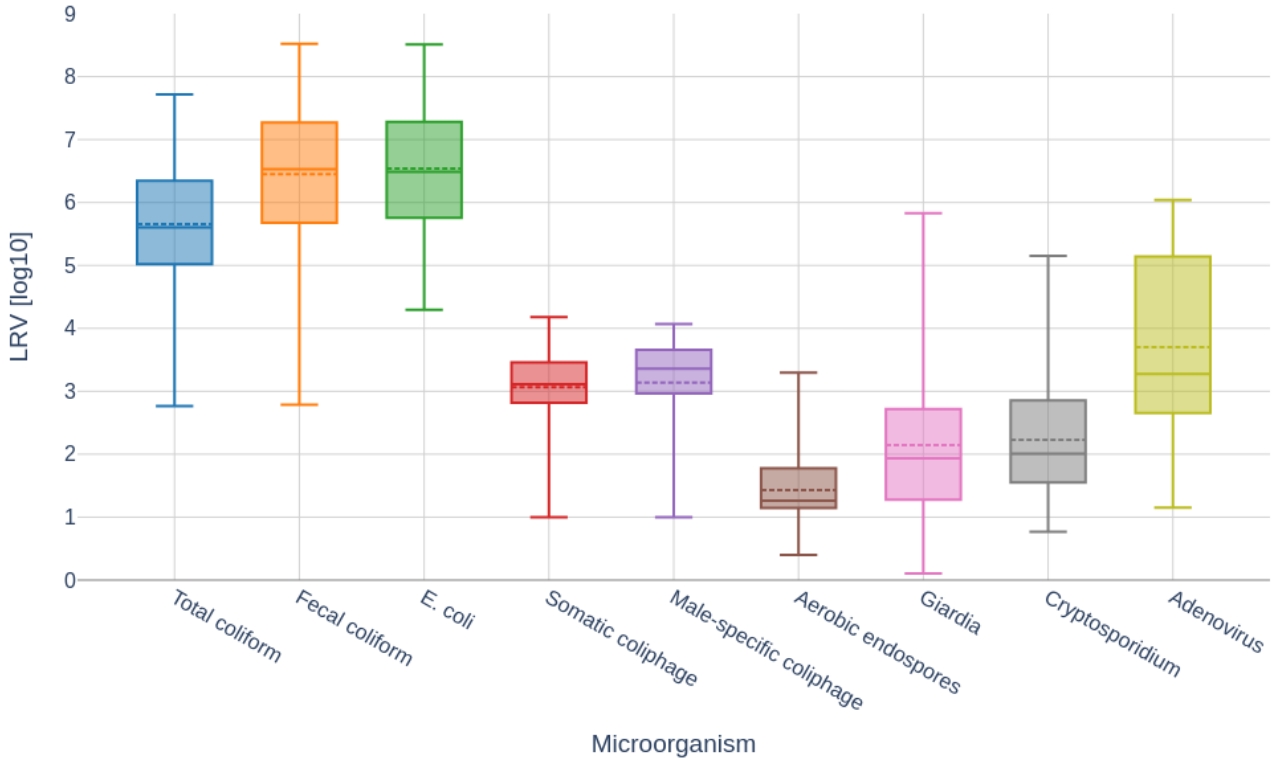

Figure 4 – LRV distribution pooled across plants for all microorganisms.

Table 6 – Log-removal value (LRV) summary statistics (all plants pooled; log10).

Microorganism	n pairs	Median LRV (log10)	Mean LRV (log10)	P25 LRV (log10)	P75 LRV (log10)
Adenovirus	46	3.3	3.7	2.7	5.1
Aerobic endospores	45	1.3	1.4	1.1	1.8
Cryptosporidium	31	2	2.2	1.6	2.8
E. coli	44	6.5	6.5	5.8	7.3
Fecal coliform	44	6.5	6.4	5.7	7.2
Giardia	46	1.9	2.1	1.3	2.7
Male-specific coliphage	45	3.4	3.1	3	3.6
Somatic coliphage	44	3.1	3.1	2.8	3.4
Total coliform	41	5.6	5.7	5	6.3

Pooled LRV patterns indicate that aerobic endospores have the lowest removal among the indicators, consistent with persistence and association with solids. Coliphages occupy an intermediate removal band. Bacterial indicators show the highest LRVs and the most compressed spread, which limits their ability to resolve pathogen removal variability when data are pooled across plants and events.

4.5 Surrogate effectiveness: pathogen LRV vs indicator LRV

Surrogacy was evaluated using event-matched comparisons of pathogen LRVs versus indicator LRVs with all plants pooled. Figures 5 to 7 show scatter plots for adenovirus, Giardia, and

Cryptosporidium, respectively. Table 7 summarizes Spearman and Pearson correlations for the same matched datasets.

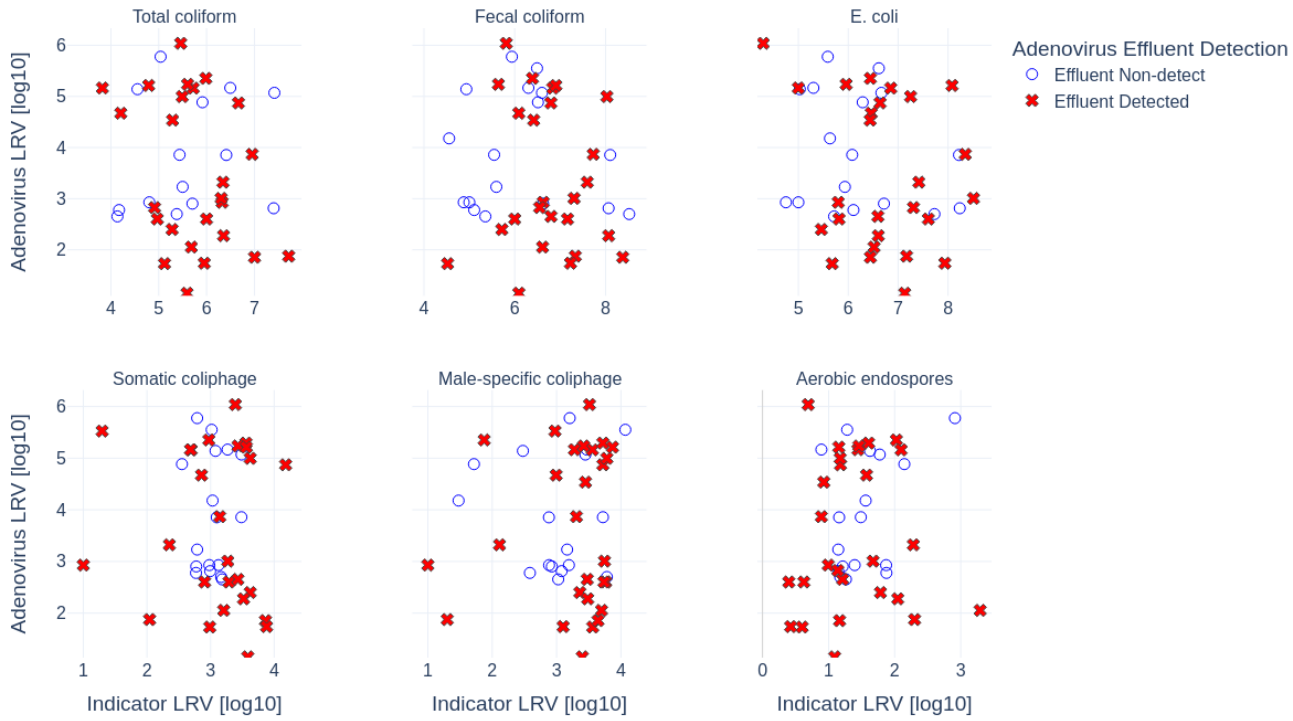

Figure 5 – Event-matched scatter: adenovirus LRV versus indicator LRVs (all plants pooled).

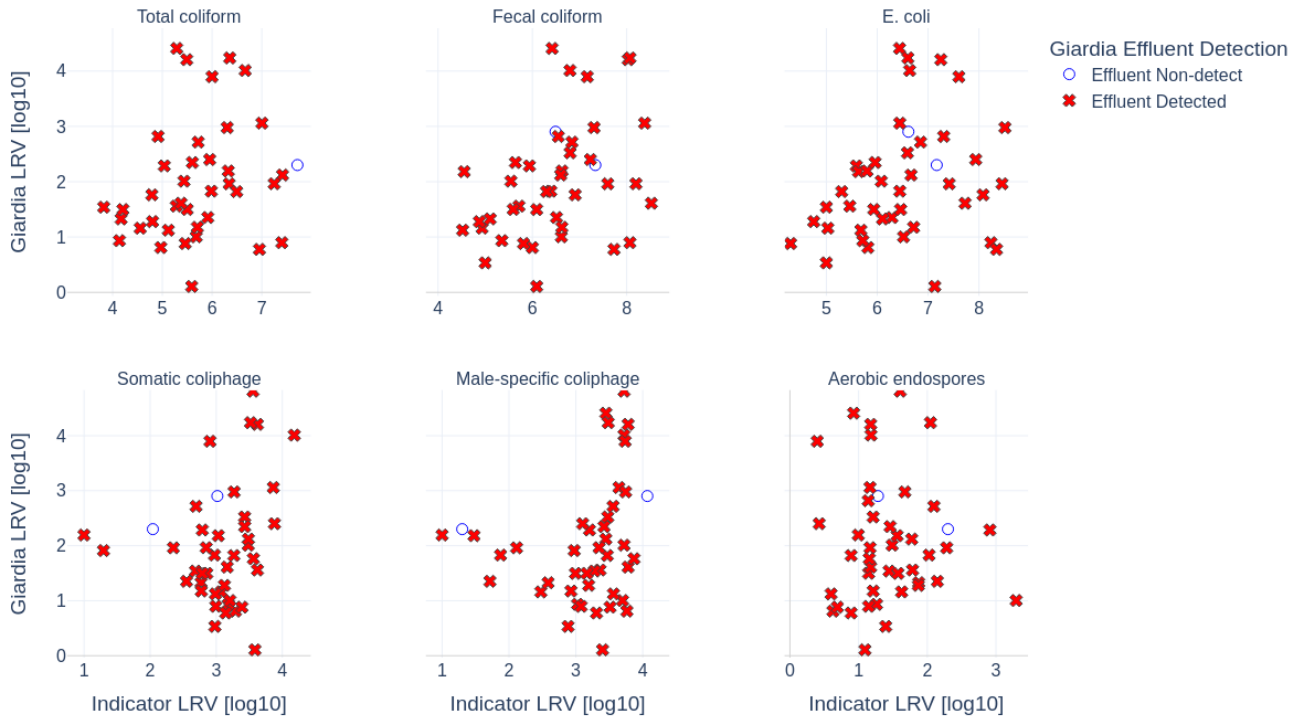

Figure 6 – Event-matched scatter: Giardia LRV versus indicator LRVs (all plants pooled).

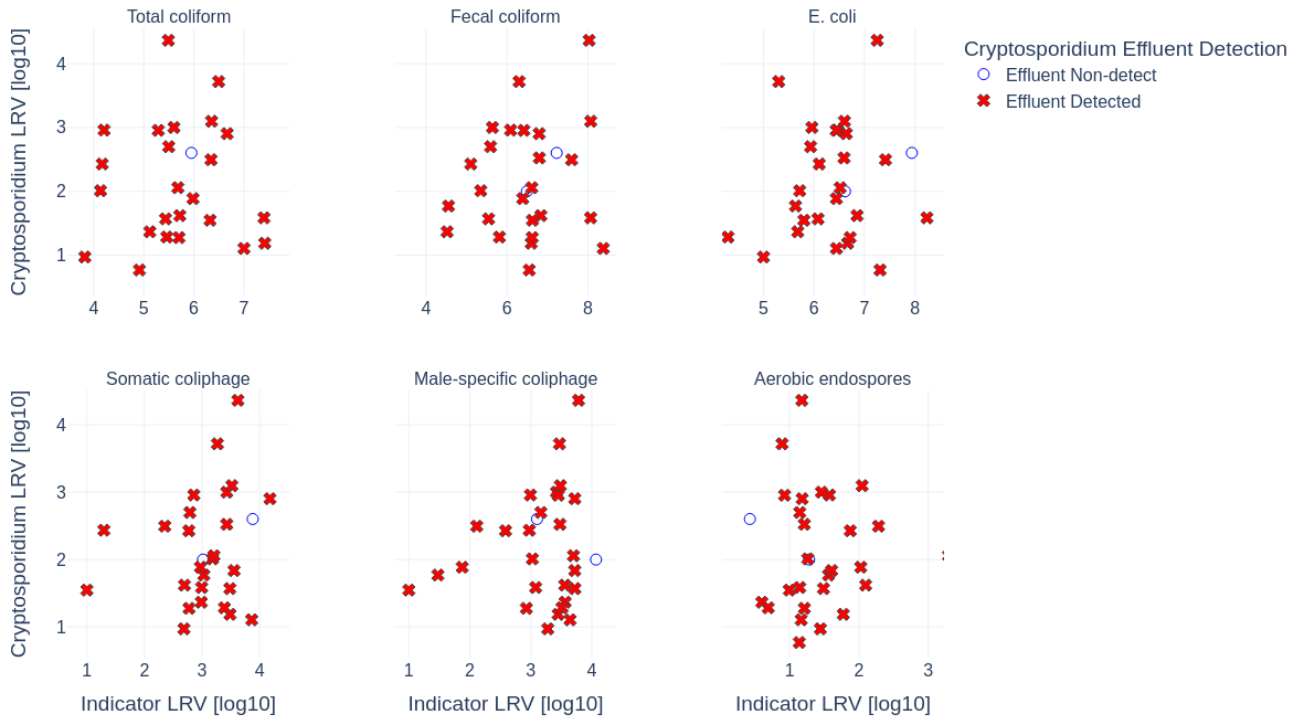

Figure 7 – Event-matched scatter: Cryptosporidium LRV versus indicator LRVs (all plants pooled).

Table 7 – Correlation between indicator LRVs and pathogen LRVs (all plants pooled; matched events).

Pathogen	Indicator	n pairs	Spearman ρ	Pearson r
Adenovirus	Aerobic endospores	44	0.187	0.125
Adenovirus	E. coli	43	-0.202	-0.238
Adenovirus	Fecal coliform	43	-0.201	-0.189
Adenovirus	Male-specific coliphage	44	0.026	0.055
Adenovirus	Somatic coliphage	43	-0.158	-0.067
Adenovirus	Total coliform	40	-0.085	-0.095
Cryptosporidium	Aerobic endospores	29	0.071	-0.008
Cryptosporidium	E. coli	28	0.077	0.142
Cryptosporidium	Fecal coliform	28	0.144	0.283
Cryptosporidium	Male-specific coliphage	29	0.068	0.155
Cryptosporidium	Somatic coliphage	28	0.231	0.186
Cryptosporidium	Total coliform	25	0.068	0.01
Giardia	Aerobic endospores	44	0.093	-0.023
Giardia	E. coli	43	0.312	0.263
Giardia	Fecal coliform	43	0.426	0.392
Giardia	Male-specific coliphage	44	0.32	0.219
Giardia	Somatic coliphage	43	0.205	0.202
Giardia	Total coliform	40	0.365	0.305

When pooled across plants, surrogate relationships are generally weak. Bacterial indicator LRVs are consistently high and compressed, producing vertical point clouds against pathogen LRVs

that vary more widely. Coliphages provide more spread in indicator LRV but still show substantial pathogen variability at similar coliphage LRV. These results support using indicators primarily for operational verification within a defined treatment context, rather than as stand-alone predictors of virus or protozoa removal across different plants and configurations.

5. Key Findings

Key findings are summarized below, integrating pooled detection prevalence (Tables 2 and 3), pooled concentration context (Tables 4 and 5), removal performance (Figure 4 and Table 6), and surrogate analyses (Figures 5 to 7 and Table 7).

- Influent detections were near-universal for bacterial indicators and coliphages, while Cryptosporidium influent prevalence was materially lower than other targets (Table 2).
- Effluent detections differed strongly by organism group: aerobic endospores remained detected in essentially all effluents, while coliphages were frequently non-detect, indicating strong reduction and/or MDL censoring (Table 3).
- Bacterial indicators achieved the highest LRVs overall (typically around 5 to 7 log10), demonstrating robust removal of bacterial targets, but with limited ability to resolve event-to-event pathogen variability (Figure 4 and Table 6).
- Protozoan removal (Giardia and Cryptosporidium) was substantially lower than bacterial indicators, indicating that fecal indicator LRVs overstate protozoa removal performance when used as surrogates (Figure 4 and Table 6).
- Adenovirus removal showed high variability relative to indicators, consistent with disinfection sensitivity and episodic performance excursions (Figure 4).
- Correlations between indicator LRVs and pathogen LRVs were generally weak when pooled across plants, indicating limited surrogate performance without additional process context (Table 7).

6. Recommendations

Recommendations are provided to strengthen performance assessment and improve the practical use of indicators for operational control and risk-based decision-making across the three plants.

- Maintain direct pathogen monitoring for protozoa and viruses where risk targets depend on these organisms; bacterial fecal indicators alone are not sufficient to infer protozoa or adenovirus removal across different treatment trains (Figures 5 to 7 and Table 7).
- Use coliphages as a potential intermediate indicator where viral surrogacy is desired, but validate performance relationships within each plant and treatment configuration rather than relying on pooled relationships (Table 7).
- For plants with disinfection-driven variability in virus removal, prioritize verification of disinfection performance during low-LRV events (e.g., UV dose delivery and UV transmittance for UV systems; disinfectant residual and contact time for chlorine systems), and link microbial excursions to operational records where possible.
- For protozoa control, focus on unit processes responsible for physical removal and particle capture (e.g., clarification, filtration/microscreens). Where available, add complementary

surrogates tied to physical removal (turbidity, particle counts, TSS, filter headloss) to support mechanistic interpretation of protozoa LRV variability.

- Further analyses recommended: (1) plant-specific surrogate assessments (correlations and conditional performance) rather than pooled; (2) seasonal/monthly stratification to identify weather- or temperature-linked performance changes; (3) censoring-aware statistics for MDL-heavy targets such as coliphages; and (4) event-based investigations linking low LRVs to operational changes, hydraulics, and unit process performance.

Appendix A. Full Tables by Plant

This appendix provides the full by-plant tables that underpin the pooled summaries presented in the main body of the report.

Appendix A.1 Detection prevalence (by plant and pooled)

Tables 8 and 9 provide influent and effluent detection prevalence by plant and pooled across all plants.

Table 8 – Influent detection prevalence by plant and pooled (all targets).

Plant	Microorganism	n	Detections	Prevalence (%)
BiSp	Adenovirus	16	16	100.0
BiSp	Aerobic endospores	14	14	100.0
BiSp	Cryptosporidium	16	11	68.8
BiSp	E. coli	14	14	100.0
BiSp	Fecal coliform	14	14	100.0
BiSp	Giardia	16	16	100.0
BiSp	Male-specific coliphage	14	14	100.0
BiSp	Somatic coliphage	13	13	100.0
BiSp	Total coliform	13	13	100.0
BrWo	Adenovirus	16	14	87.5
BrWo	Aerobic endospores	15	15	100.0
BrWo	Cryptosporidium	15	10	66.7
BrWo	E. coli	15	15	100.0
BrWo	Fecal coliform	15	15	100.0
BrWo	Giardia	15	14	93.3
BrWo	Male-specific coliphage	15	15	100.0
BrWo	Somatic coliphage	15	15	100.0
BrWo	Total coliform	14	14	100.0
NoWe	Adenovirus	16	16	100.0
NoWe	Aerobic endospores	16	16	100.0
NoWe	Cryptosporidium	16	10	62.5
NoWe	E. coli	16	16	100.0
NoWe	Fecal coliform	16	16	100.0
NoWe	Giardia	16	16	100.0
NoWe	Male-specific coliphage	16	16	100.0
NoWe	Somatic coliphage	16	16	100.0
NoWe	Total coliform	15	15	100.0
All plants pooled	Adenovirus	48	46	95.8
All plants pooled	Aerobic endospores	45	45	100.0
All plants pooled	Cryptosporidium	47	31	66.0
All plants pooled	E. coli	45	45	100.0
All plants pooled	Fecal coliform	45	45	100.0
All plants pooled	Giardia	47	46	97.9
All plants pooled	Male-specific coliphage	45	45	100.0
All plants pooled	Somatic coliphage	44	44	100.0
All plants pooled	Total coliform	42	42	100.0

Table 9 – Effluent detection prevalence by plant and pooled (all targets).

Plant	Microorganism	n	Detections	Prevalence (%)
BiSp	Adenovirus	16	13	81.2
BiSp	Aerobic endospores	14	14	100.0
BiSp	Cryptosporidium	16	12	75.0
BiSp	E. coli	13	8	61.5
BiSp	Fecal coliform	13	10	76.9
BiSp	Giardia	16	14	87.5
BiSp	Male-specific coliphage	15	1	6.7
BiSp	Somatic coliphage	15	2	13.3
BiSp	Total coliform	12	11	91.7
BrWo	Adenovirus	16	6	37.5
BrWo	Aerobic endospores	15	15	100.0

BrWo	Cryptosporidium	15	14	93.3
BrWo	E. coli	15	7	46.7
BrWo	Fecal coliform	15	7	46.7
BrWo	Giardia	15	15	100.0
BrWo	Male-specific coliphage	15	1	6.7
BrWo	Somatic coliphage	15	2	13.3
BrWo	Total coliform	14	13	92.9
NoWe	Adenovirus	16	9	56.2
NoWe	Aerobic endospores	16	16	100.0
NoWe	Cryptosporidium	16	13	81.2
NoWe	E. coli	16	16	100.0
NoWe	Fecal coliform	16	16	100.0
NoWe	Giardia	16	16	100.0
NoWe	Male-specific coliphage	16	3	18.8
NoWe	Somatic coliphage	16	5	31.2
NoWe	Total coliform	15	15	100.0
All plants pooled	Adenovirus	48	28	58.3
All plants pooled	Aerobic endospores	45	45	100.0
All plants pooled	Cryptosporidium	47	39	83.0
All plants pooled	E. coli	44	31	70.5
All plants pooled	Fecal coliform	44	33	75.0
All plants pooled	Giardia	47	45	95.7
All plants pooled	Male-specific coliphage	46	5	10.9
All plants pooled	Somatic coliphage	46	9	19.6
All plants pooled	Total coliform	41	39	95.1

Appendix A.2 Concentration statistics (by plant and pooled)

Tables 10 and 11 provide combined concentration statistics (median, mean, and interquartile range) for influent and effluent, by plant and pooled.

Table 10 – Influent concentrations summary by plant and pooled (original units).

Plant	Microorganism	n	Median	Mean	P25	P75
BiSp	Adenovirus	16	64	3,047	16	6,087
BiSp	Aerobic endospores	14	158	629	88	391
BiSp	Cryptosporidium	16	40	736	0	1,182
BiSp	E. coli	14	9,530,000	95,076,429	3,085,000	45,125,000
BiSp	Fecal coliform	14	23,750,000	179,161,429	7,400,000	238,750,000
BiSp	Giardia	16	2,400	4,576	198	6,958
BiSp	Male-specific coliphage	14	5,125	4,985	2,975	5,738
BiSp	Somatic coliphage	13	1,865	2,548	1,300	3,600
BiSp	Total coliform	13	145,000,000	631,692,308	47,100,000	461,100,000
BrWo	Adenovirus	16	18	904	15	479
BrWo	Aerobic endospores	15	265	709	98	1,075
BrWo	Cryptosporidium	15	10	21	0	20
BrWo	E. coli	15	5,200,000	49,584,667	3,355,000	39,715,000
BrWo	Fecal coliform	15	8,500,000	106,242,667	5,130,000	83,340,000
BrWo	Giardia	15	410	971	105	1,245
BrWo	Male-specific coliphage	15	1,555	2,135	805	2,902
BrWo	Somatic coliphage	15	985	2,171	660	1,780
BrWo	Total coliform	14	32,550,000	261,225,714	19,337,500	180,675,000
NoWe	Adenovirus	16	3,025	9,672	262	11,471
NoWe	Aerobic endospores	16	218	383	140	310
NoWe	Cryptosporidium	16	10	35	0	50
NoWe	E. coli	16	9,850,000	101,100,000	4,737,500	59,202,500
NoWe	Fecal coliform	16	19,850,000	272,849,375	6,035,000	124,175,000
NoWe	Giardia	16	270	619	150	658
NoWe	Male-specific coliphage	16	1,432	2,082	835	2,789
NoWe	Somatic coliphage	16	1,458	1,873	908	2,784
NoWe	Total coliform	15	101,700,000	403,934,000	21,100,000	318,600,000

All plants pooled	Adenovirus	48	107	4,541	17	5,819
All plants pooled	Aerobic endospores	45	210	568	110	445
All plants pooled	Cryptosporidium	47	10	269	0	50
All plants pooled	E. coli	45	7,120,000	82,054,222	4,000,000	53,400,000
All plants pooled	Fecal coliform	45	17,900,000	188,166,444	6,200,000	137,000,000
All plants pooled	Giardia	47	410	2,078	120	2,510
All plants pooled	Male-specific coliphage	45	2,310	3,003	1,060	5,000
All plants pooled	Somatic coliphage	44	1,430	2,174	906	2,784
All plants pooled	Total coliform	42	83,900,000	426,861,190	25,732,500	327,650,000

Table 11 – Effluent concentrations summary by plant and pooled (original units).

Plant	Microorganism	n	Median	Mean	P25	P75
BiSp	Adenovirus	16	0	0	0	0
BiSp	Aerobic endospores	14	7	24	3	11
BiSp	Cryptosporidium	16	0	4	0	2
BiSp	E. coli	13	1	22	1	4
BiSp	Fecal coliform	13	3	18	1	15
BiSp	Giardia	16	1	7	0	4
BiSp	Male-specific coliphage	15	1	2	1	1
BiSp	Somatic coliphage	15	1	5	1	1
BiSp	Total coliform	12	406	853	132	1,446
BrWo	Adenovirus	16	0	0	0	0
BrWo	Aerobic endospores	15	9	19	6	18
BrWo	Cryptosporidium	15	0	0	0	1
BrWo	E. coli	15	1	14	1	1
BrWo	Fecal coliform	15	1	18	1	3
BrWo	Giardia	15	5	17	3	17
BrWo	Male-specific coliphage	15	1	16	1	1
BrWo	Somatic coliphage	15	1	15	1	1
BrWo	Total coliform	14	61	208	14	92
NoWe	Adenovirus	16	0	45	0	0
NoWe	Aerobic endospores	16	9	16	5	20
NoWe	Cryptosporidium	16	0	1	0	1
NoWe	E. coli	16	9	14	4	15
NoWe	Fecal coliform	16	49	113	27	122
NoWe	Giardia	16	10	20	4	24
NoWe	Male-specific coliphage	16	1	2	1	1
NoWe	Somatic coliphage	16	1	1	1	1
NoWe	Total coliform	15	816	963	500	1,120
All plants pooled	Adenovirus	48	0	15	0	0
All plants pooled	Aerobic endospores	45	8	19	4	18
All plants pooled	Cryptosporidium	47	0	2	0	1
All plants pooled	E. coli	44	2	16	1	9
All plants pooled	Fecal coliform	44	7	52	1	46
All plants pooled	Giardia	47	4	15	3	20
All plants pooled	Male-specific coliphage	46	1	6	1	1
All plants pooled	Somatic coliphage	46	1	7	1	1
All plants pooled	Total coliform	41	345	673	68	1,120

Appendix A.3 LRV statistics (by plant and pooled)

Table 12 provides combined LRV summary statistics (median, mean, and interquartile range) by plant and pooled across all plants.

Table 12 – Log-removal value (LRV) summary statistics by plant and pooled (log10).

Plant	Microorganism	n pairs	Median LRV (log10)	Mean LRV (log10)	P25 LRV (log10)	P75 LRV (log10)
BiSp	Adenovirus	16	2.9	3.5	2.4	5
BiSp	Aerobic endospores	14	1.2	1.5	1	1.8
BiSp	Cryptosporidium	11	2.4	2.7	1.9	3.4
BiSp	E. coli	13	6.6	6.7	6.4	7.3
BiSp	Fecal coliform	13	6.6	7	6.4	7.3
BiSp	Giardia	16	2.9	3	1.9	4.2
BiSp	Male-specific coliphage	14	3.7	3.4	3.5	3.7
BiSp	Somatic coliphage	13	3.3	3.1	3	3.6
BiSp	Total coliform	12	5.8	6	5.3	6.4
BrWo	Adenovirus	14	3.1	3.6	2.9	4.6
BrWo	Aerobic endospores	15	1.2	1.5	1.2	1.8
BrWo	Cryptosporidium	10	1.8	2	1.6	2.5
BrWo	E. coli	15	6.7	6.8	6.2	7.6
BrWo	Fecal coliform	15	6.8	6.8	6.5	7.8
BrWo	Giardia	14	1.9	1.9	1.3	2.2
BrWo	Male-specific coliphage	15	3.2	3	2.9	3.5
BrWo	Somatic coliphage	15	3	2.9	2.8	3.1
BrWo	Total coliform	14	6.2	6	5.6	6.6
NoWe	Adenovirus	16	4.4	4	2.7	5.2
NoWe	Aerobic endospores	16	1.5	1.4	1	1.6
NoWe	Cryptosporidium	10	1.9	2	1.4	2.6
NoWe	E. coli	16	6	6.2	5.6	6.6
NoWe	Fecal coliform	16	5.7	5.7	5.1	6.2
NoWe	Giardia	16	1.4	1.5	1.1	2.1
NoWe	Male-specific coliphage	16	3.2	3	2.9	3.4
NoWe	Somatic coliphage	16	3.1	3.2	2.8	3.4
NoWe	Total coliform	15	5.1	5.1	4.4	5.6
All plants pooled	Adenovirus	46	3.3	3.7	2.7	5.1
All plants pooled	Aerobic endospores	45	1.3	1.4	1.1	1.8
All plants pooled	Cryptosporidium	31	2	2.2	1.6	2.8
All plants pooled	E. coli	44	6.5	6.5	5.8	7.3
All plants pooled	Fecal coliform	44	6.5	6.4	5.7	7.2
All plants pooled	Giardia	46	1.9	2.1	1.3	2.7
All plants pooled	Male-specific coliphage	45	3.4	3.1	3	3.6
All plants pooled	Somatic coliphage	44	3.1	3.1	2.8	3.4
All plants pooled	Total coliform	41	5.6	5.7	5	6.3